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POP-TAG Study

Perimetric **O**utcomes of **P**ortable **T**ablet-based **A**ssessment in **G**laucoma

Perimetric Outcomes of Tablet-Based Perimetry in Patients with Glaucoma: A Systematic Review and Meta-Analysis

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Title	
Title of Your Review	Perimetric Outcomes of Tablet-Based Perimetry in Patients with Glaucoma: A Systematic Review and Meta-Analysis
Registration	
PROSPERO	**PROSPERO**: CRD420251009468
Cloud Storage: Protocol Documents, Literature Searches and Manuals	**Google Drive:**https://drive.google.com/drive/folders/1ajhZME117FfhNGqzBSI-mXy8TyFZgdnk?usp=share_link
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CRediT (Contribution Roles Taxonomy)	***Andres Inzunza, MD***- *andres_inzunza@hms.harvard.edu ; ainzunzamartindelcampo@meei.harvard.edu* ***First Author*** ● *Conceptualization* ● *Data Curation* ● *Methodology* ● *Investigation* ● *Formal Analysis*

	• *Visualization* • *Writing – Original Draft Preparation* • *Writing – Review & Editing* ***Gabriela Petersen, MD-*** *gabrielapetersenmd@gmail.com* • *Conceptualization* • *Data Curation* • *Methodology* • *Investigation* • *Visualization* • *Writing – Original Draft Preparation* • *Writing – Review & Editing* ***Edgar Alejandro Moreno-Díaz, MD-*** alexmorenodz@gmail.com • *Data Curation* • *Investigation* • *Formal Analysis* • *Writing – Original Draft Preparation* • *Writing – Review & Editing* ***Deborah Goss, MLS,*** deborah_goss@meei.harvard.edu ***Consultant Medical Librarian*** • *Search Strategy* • *Methodology* • *Writing – Original Draft Preparation* • *Writing – Review & Editing* ***David Friedman, MD, MPH, PhD*** David_Friedman@MEEI.HARVARD.EDU ***Principal Investigator*** • *Conceptualization* • *Methodology* • *Validation* • *Supervision* • *Writing – Review & Editing*

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Financial Support & Conflicts of Interest	
Financial Support	Two authors (AI and EAMD) received a financial grant from the Mexican National Council of Science and Technology (CONAHCYT) to support their graduate education. D.S.F. is a consultant for Théa Pharmaceuticals and Life Biosciences and has received honoraria for lecturing. D.S.F. has received research support from Genentech and Perivision. No private or public entity played a role in the design or conduct of the study; collection, management, analysis, or interpretation of the data; preparation, review, or approval of the manuscript; or the decision to submit the manuscript for publication.

Review Rationale	
Rationale	Glaucoma is the leading cause of irreversible vision loss worldwide. Due to its slow and progressive asymptomatic nature, glaucomatous vision loss often remains undetected for years. Serial visual field (VF) testing is crucial for diagnosing and monitoring disease progression. Despite the importance of serial visual field testing, there is no universally accepted method for evaluating VF progression in glaucoma. The Humphrey Field Analyzer (HFA) using the Swedish Interactive Threshold Algorithm (SITA) Standard (Carl Zeiss Meditec, Inc., Dublin, CA, USA) is one of the most widely used perimetry techniques in clinical practice. Despite its widespread adoption, HFA testing is time-consuming and costly.

	Recently, tablet or computer-based portable VF testing has emerged as a potential alternative to traditional clinic-based perimetry. Tablet-based, computer-based, and other portable automated perimeters may allow for more frequent assessments, potentially allowing for earlier detection of VF progression. However, despite recent technological advancements, studies evaluating these devices have produced mixed results due to heterogeneous study populations and methodologies. To address this gap, we will conduct a systematic review and meta-analysis to answer this question.
Review Objectives	
Research Question	**Research Question:** In adult patients with glaucoma, suspected glaucoma, or ocular hypertension, how do perimetric outcomes obtained from tablet-based portable perimetry devices compare to those obtained with standard perimetry? ● **P (Population):** Adults diagnosed with glaucoma, suspected glaucoma, or ocular hypertension undergoing visual field testing. ● **I (Intervention)** Visual function assessment using tablet-based portable perimetry devices. ● **C (Comparison):** Standard automated perimetry using the Humphrey Field Analyzer (24-2 Swedish Interactive Threshold Algorithm, Standard program). ● **O (Outcomes):** o Mean deviation (MD), pattern standard deviation (PSD), point sensitivity, and reliability parameters (false positives, false negatives, fixation losses).
Methods	

Guidelines	**Conducting**: This systematic review will be conducted according to the **Institute of Medicine, Finding What Works in Health Care: Standards for Systematic Reviews** *Institute of Medicine. 2011. Finding What Works in Health Care: Standards for Systematic Reviews. Washington, DC: The National Academies Press. https://doi.org/10.17226/13059 .* Any changes to proposed methods will be publicly recorded and updated. **Reporting**: Results will be reported using PRISMA guidelines.
Eligibility	
Inclusion	● Studies involving adults diagnosed with glaucoma, suspected glaucoma, or ocular hypertension. ● Studies assessing visual field outcomes using tablet or pc-based portable perimetry devices. ● Studies comparing outcomes to the Humphrey Field Analyzer (24-2 Swedish Interactive Threshold Algorithm, Standard program). ● Studies reporting at least one primary outcome of interest. ● Studies in English or Spanish language
Exclusion	● Studies involving pediatric populations. ● Patients with other ocular conditions unrelated to glaucoma. ● Studies not comparing tablet-based perimeter performance or comparison with the Humphrey Field Analyzer. ● Case reports, letters to the editor, expert opinions, and narrative reviews. ● Non-English or Spanish articles.

	• Studies with duplicate data already included in another publication.
Outcomes and Priorities	
Primary Outcomes	**Primary Outcomes:** Mean deviation (MD), pattern standard deviation (PSD), point sensitivity, and reliability parameters (false positives, false negatives, fixation losses). **Secondary Outcomes:** Patient-reported measures such as ease of use, comfort, test duration, and acceptability.

Methods

Databases	• EMBASE • PUBMED/Medline • Web of Science • Cochrane Library
Search Strategy	
Search Strategy	**SEARCH STRATEGY:** A systematic review of the literature was designed by the team and executed by a medical librarian (DG). It was documented according to the guidelines of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA).[1] A search of the literature was performed on Ovid MEDLINE (1946-), Embase.com (1947-), Web of Science Core Collection (1900-) and Cochrane Library (Wiley 1996-) on March 11, 2025, using controlled vocabulary and keywords focused on the concepts of visual field outcomes using standard automated

	perimetry versus tablet or PC -based portable perimetry devices. No filters for language, study design, date of publication, or country of origin were applied. All references were exported into Endnote 7.8 (Thomson Reuters, 1988-2016) for deduplication and then to Covidence (Veritas Health Innovation) for further deduplication, study screening, selection, and data extraction. 1. BMJ. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71 **Ovid Medline 3.11.2025** Exp Glaucoma/ OR glaucoma.ti,ab,kw. OR *Ocular Hypertension/ 88,435 exp Visual Field Tests/ OR “standard automat* perimetr*”.ti,ab,kw. OR “Visual Field Test*”.ti,ab,kw. OR “Visual field exam*”.ti,ab,kw. OR Campimetr*.ti,ab,kw. OR “Tangent screen exam*”.ti,ab,kw OR “humphrey field analyzer”.ti,ab,kw. OR “automated static perimetr*”.ti,ab,kw. 13,234 “Melbourne Rapid Field*”.ti,ab,kw. OR “Smart Visual Function Analyzer”.ti,ab,kw. OR portable.ti,ab,kw OR “Online circular contrast perimetry”.ti,ab,kw. OR tablet.ti,ab,kw. OR smartphone.ti,ab,kw. OR laptop.ti,ab,kw. OR PC.ti,ab,kw. OR “Virtual reality”.ti,ab,kw. OR “Head-mounted”.ti,ab,kw. OR ipad.ti,ab,kw. 218,893 1-3=125 **Embase 3.11.2025**

	'intraocular hypertension'/exp OR 'glaucoma'/exp OR glaucoma:ti,ab,kw 137,561 'perimetry'/exp OR ‘standard automat* perimetr*’:ti,ab,kw OR ‘Visual Field Test*’:ti,ab,kw OR ‘Visual field exam*’:ti,ab,kw OR Campimetr*:ti,ab,kw OR ‘Tangent screen exam*’:ti,ab,kw OR ‘humphrey field analyzer’:ti,ab,kw OR ‘automated static perimetr*’:ti,ab,kw 20,369 ‘Melbourne Rapid Field*’:ti,ab,kw OR ‘Smart Visual Function Analyzer’:ti,ab,kw OR portable:ti,ab,kw OR ‘Online circular contrast perimetry’:ti,ab,kw OR tablet:ti,ab,kw OR smartphone:ti,ab,kw OR laptop:ti,ab,kw OR PC:ti,ab,kw OR ‘Virtual reality’:ti,ab,kw OR ‘Head-mounted’:ti,ab,kw OR ipad:ti,ab,kw 311,737 1-3=222 **Web of Science (Core collection) 3.11.2025** TS=(“intraocular hypertension” OR Glaucoma) 84,805 TS=(Perimetry OR “standard automat* perimetr*” OR “Visual Field Test*” OR “Visual field exam*” OR Campimetr* OR “Tangent screen exam*” OR “humphrey field analyzer” OR “automated static perimetr*”) 11,045 TS=(“Melbourne Rapid Field*” OR “Smart Visual Function Analyzer” OR portable OR “Online circular contrast perimetry” OR tablet OR smartphone OR laptop OR PC OR “Virtual reality” OR “Head-mounted” OR ipad)

	578,875 1-3=148 **Cochrane Library 3.11.2025** Searched Ti,ab,kw (“intraocular hypertension” OR Glaucoma) AND (Perimetry OR “standard automat* perimetr*” OR “Visual Field Test” OR “Visual field exam*” OR Campimetr* OR “Tangent screen exam*” OR “humphrey field analyzer” OR “automated static perimetr*”) AND (“Melbourne Rapid Field” OR “Smart Visual Function Analyzer” OR portable OR “Online circular contrast perimetry” OR tablet OR smartphone OR laptop OR PC OR “Virtual reality” OR “Head-mounted” OR ipad) 1-3 = 32 trials The search produced 527 studies before deduplication, and 289 after deduplication.
Filters	**NO FILTER** Following the Harvard Medical School Countway Library of Medicine Guide for Systematic Reviews and Meta-Analyses, it was decided not to use filters for this study.
Data Management	
Management and Extraction	• Titles and abstracts of all identified studies will be screened independently by three reviewers (AI, GP, EAMD) using **COVIDENCE®** (Veritas Health Innovation, Melbourne, Australia). • Full-text articles will be retrieved for studies deemed potentially eligible, based on the title/abstract screening.

	● Discrepancies between the reviewers will be resolved through discussion. If needed, a third reviewer will be consulted. ● Studies meeting inclusion criteria will undergo a detailed full-text review. ● Reviewers will extract data from eligible studies using the templates within Covidence. All voting will be concealed from other reviewers. ● Once data is extracted, reviewers will synthesize evidence into data management tables. ● Extracted variables will be coded numerically for consistency. For binary variables, coding will follow a 0/1 scheme. ● Categories will be created to classify descriptive information. ● A codebook will be created and used to ensure consistency across all coders. ● The detailed search strategy, including database sources and keywords, will be fully documented and included as an appendix. ● Search results will be recorded, including the total number of studies identified, duplicates removed, and studies excluded at each stage of screening. ● Record keeping of search results, duplicate removal, and exclusions will be conducted in Covidence to reduce human error. ● Point estimates and measures of dispersion will be collected and formatted in a standardized manner. For datasets where the most common point estimate is the mean and the most common measure of dispersion is the standard deviation (SD), alternative forms of data reporting, such as the median for point estimates and the interquartile range (IQR) for dispersion, will be converted using a validated and published conversion tool. The conversion will be performed using the Meta-Converter tool【Abbas et al., 2024】, which facilitates statistical data conversion for systematic reviews with meta-analysis (https://meta-converter.com/). ● **Reference**: Abbas, A., Hefnawy, M.T. & Negida, A. Meta-analysis accelerator: a comprehensive tool

	for statistical data conversion in systematic reviews with meta-analysis. *BMC Medical Research Methodology* 24, 243 (2024). https://doi.org/10.1186/s12874-024-02356-6
Bias	
Risk of Bias Assessment	• Two reviewers will independently assess included articles for Risk of Bias using the ROBINS-1 V2 tool for non-randomized studies. • Selection Bias, Publication Bias, and Measurement Bias will be assessed.
Analyses	
Meta-Analysis • If sufficient data are gathered and there is reasonable homogeneity in data presentation, a meta-analysis will be conducted. • Weighted averages and pooled estimates will be calculated for key outcomes. • A random-effects model was defined a priori, as the expected point estimate is anticipated to vary among studies beyond what would be expected due to sampling error. • Pre-specified subgroup analyses will be conducted based on population characteristics. **4.2. Qualitative Analysis** • Outcomes that cannot be categorized or pooled quantitatively will be summarized using a narrative synthesis approach. • The narrative synthesis will provide a structured summary of study findings.	
Existing Meta-Biases	
Meta-Biases	The potential for publication bias will be further explored by funnel plots if ≥10 studies are available. **Reference**: Sterne J A C, Sutton A J, Ioannidis J P A, Terrin N, Jones D R, Lau J et al. Recommendations for examining and interpreting funnel plot asymmetry in

	meta-analyses of randomised controlled trials*BMJ* 2011; 343 :d4002 doi:10.1136/bmj.d4002
Reproducibility	
Reproducibility:	**Search Strategy Documentation:** ● The detailed search strategy, including database sources and keywords will be fully documented and included as an appendix. ● Search results will be recorded, including the total number of studies identified, duplicates removed, and studies excluded at each stage of screening. ● Record keeping of search results, duplicate removal, and exclusions will be conducted in Covidence to reduce human error.
Study Selection and Data Extraction:	● A standardized protocol for study selection and data extraction will be followed. ● A record of inclusion and exclusion decisions for each study will be maintained.
Statistical Transparency:	● The statistical analysis code and DO FILE (Stata 18) will be publicly available. ● Specifications for meta-analysis models and inclusion criteria for subgroup analysis will be defined a priori and shared online.